

Table 2: Global development on quantum computers by tech giants

Company	Type	Working pattern
Nvidia	Quantum software + GPU/HPC	CUDA-Q, GPU-accelerated quantum simulation, hybrid quantum-classical computing
IBM	Superconducting QPU	Qiskit and quantum processors
Google	Superconducting QPU	Cirq and quantum processors
IonQ	Trapped-ion QPU	Trapped-ion quantum computers
Quantinuum	Trapped-ion QPU	Trapped-ion quantum computers
QuEra	Neutral-atom QPU	Neutral-atom quantum computers
PsiQuantum	Photonic QPU	Photonic quantum computers
D-Wave	Quantum annealing	Quantum annealing systems
Xanadu	Photonic	Photonic quantum hardware + PennyLane
Intel	Silicon-spin	Silicon-spin quantum hardware
Microsoft	Topological	Topological/Majorana approach

doing at present in this domain.

NVIDIA CUDA-Q

Nvidia, a leader in the AI world, is now also speeding up the arrival of the quantum computing era with the release of the NVIDIA CUDA-Q (<https://developer.nvidia.com/cuda-q>), a powerful open source platform for the development, simulation and implementation of quantum algorithms. The platform is popularly being referred to as the foundation for accelerated quantum-based supercomputing.

The key features of CUDA-Q are:

- CPU-based quantum simulation
- GPU-accelerated simulation
- HPC integration
- Hamiltonian simulation
- Hybrid quantum-classical computing
- Multi-GPU simulation
- Multiple backend support
- Python and C++ support
- QAOA and VQE
- Quantum algorithm development
- Quantum circuit programming
- Quantum hardware integration
- Quantum machine learning
- Quantum noise simulation

This open source portal can be

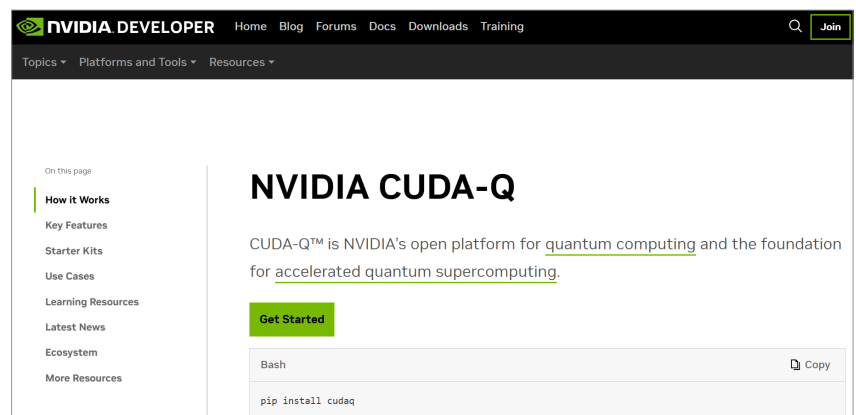


Figure 1: NVIDIA CUDA-Q framework

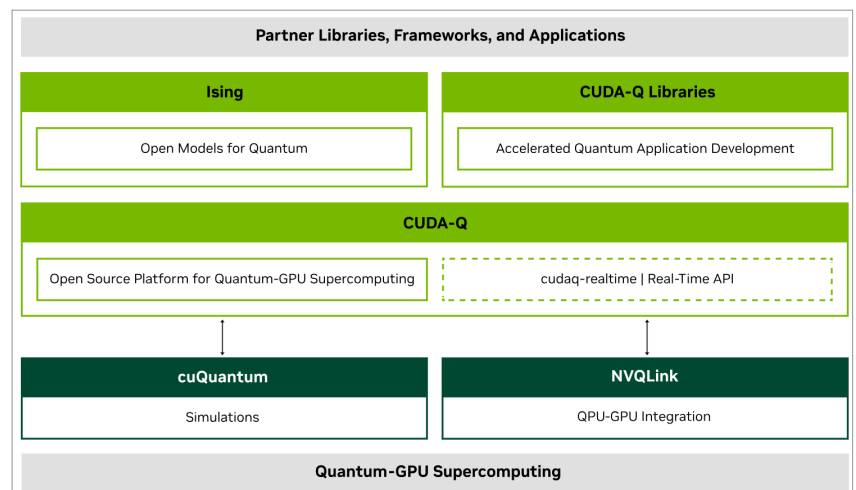


Figure 2: Technology ecosystem of NVIDIA CUDA-Q including partner libraries, frameworks and applications